NEET (UG) 2027 — Model Question Paper 1

Duration: 3 hours · Maximum marks: 720 · 180 questions (Physics 45, Chemistry 45, Botany 45, Zoology 45)

1. All questions are compulsory multiple-choice with one correct option.
2. +4 for each correct answer, −1 for each wrong answer, 0 if unattempted.
3. Take g = 10 m/s² unless stated otherwise.
4. Answer key is at the end of the paper.

Physics

1. The SI unit of luminous intensity is

(1) lumen **(2)** candela
(3) lux **(4)** watt

2. The number of significant figures in 0.00450 is

(1) 3 **(2)** 5
(3) 2 **(4)** 6

3. The area under a velocity–time graph gives

(1) acceleration **(2)** velocity
(3) displacement **(4)** jerk

4. A body dropped from rest falls through a distance in the 3rd second of (g = 10 m/s²)

(1) 45 m **(2)** 15 m
(3) 30 m **(4)** 25 m

5. Two forces of 3 N and 4 N act at right angles. The resultant is

(1) 7 N **(2)** 5 N
(3) 1 N **(4)** 12 N

6. A stone is thrown horizontally from a 45 m high cliff (g = 10 m/s²). It reaches the ground after

(1) 3 s **(2)** 4.5 s
(3) 9 s **(4)** 1.5 s

7. Newton's first law of motion defines

(1) momentum **(2)** energy
(3) inertia **(4)** power

8. A 60 kg person stands in a lift accelerating upward at 2 m/s² (g = 10 m/s²). The apparent weight is

(1) 600 N **(2)** 480 N
(3) 660 N **(4)** 720 N

9. If μ is the coefficient of static friction, the angle of repose is

(1) $\sin^{-1}\mu$ **(2)** $\tan^{-1}\mu$
(3) $\cos^{-1}\mu$ **(4)** $\cot^{-1}\mu$

10. The work done by the net force on a body equals the change in its

(1) kinetic energy **(2)** potential energy
(3) momentum **(4)** total energy

11. In a 1-D elastic collision of two equal masses with one at rest, the two bodies

(1) stick together **(2)** both stop

(3) exchange velocities **(4)** move together

12. The moment of inertia of a thin ring about a diameter is

(1) MR^2 **(2)** $2MR^2$

(3) $MR^2/4$ **(4)** $MR^2/2$

13. The radius of gyration of a uniform disc about its central axis perpendicular to its plane is

(1) R **(2)** $R/\sqrt{2}$

(3) R/2 **(4)** $\sqrt{2}$ R

14. The weight of a body at the centre of the Earth is

(1) zero **(2)** maximum

(3) same as on the surface **(4)** infinite

15. The period of a geostationary satellite is

(1) 12 h **(2)** 365 days

(3) 24 h **(4)** 1 h

16. Stress has the same dimensions as

(1) force **(2)** strain

(3) energy **(4)** pressure

17. A hydraulic lift works on

(1) Archimedes' principle **(2)** Pascal's law

(3) Bernoulli's theorem **(4)** Stokes' law

18. The SI unit of coefficient of viscosity is

(1) Pa s **(2)** N m

(3) N s **(4)** Pa m

19. Water has its maximum density at

(1) 0 °C **(2)** 100 °C

(3) 4 °C **(4)** −4 °C

20. The ratio Cp/Cv for a rigid diatomic ideal gas is

(1) 5/3 **(2)** 4/3

(3) 9/7 **(4)** 7/5

21. In an isothermal process for an ideal gas

(1) $Q = 0$ **(2)** $\Delta U = 0$

(3) $W = 0$ **(4)** $\Delta T \neq 0$

22. According to the Kelvin–Planck statement, it is impossible to

(1) convert heat completely into work in a cycle **(2)** convert work into heat

(3) transfer heat from hot to cold **(4)** compress a gas

23. The number of translational degrees of freedom of a monatomic gas molecule is

(1) 1 **(2)** 5

(3) 3 **(4)** 6

24. The time period of a mass m on a spring of constant k is

(1) $2\pi\sqrt{(k/m)}$ **(2)** $\sqrt{(m/k)}$

(3) $\pi\sqrt{(m/k)}$ **(4)** $2\pi\sqrt{(m/k)}$

25. The speed of sound in an ideal gas is proportional to

(1) T **(2)** $\sqrt{T}$

(3) $1/T$ **(4)** T^2

26. When a sound source moves towards a stationary listener, the observed frequency

(1) increases **(2)** decreases

(3) is unchanged **(4)** becomes zero

27. When two charges are placed in a medium of dielectric constant K, the force between them

(1) becomes K times **(2)** is unchanged

(3) becomes 1/K times **(4)** becomes K^2 times

28. Electric field lines

(1) form closed loops **(2)** start on negative charges

(3) can intersect **(4)** never intersect each other

29. The electric potential due to a point charge varies with distance r as

(1) $1/r^2$ **(2)** $1/r$

(3) r **(4)** r^2

30. Energy stored in a capacitor C charged to potential V is

(1) $\frac{1}{2}CV^2$ **(2)** CV^2

(3) $\frac{1}{2}CV$ **(4)** C^2V

31. The resistivity of a metal wire depends on

(1) its length **(2)** its area

(3) its material and temperature **(4)** the current

32. Kirchhoff's junction law is based on conservation of

(1) energy **(2)** momentum

(3) mass **(4)** charge

33. A potentiometer is preferred over a voltmeter for measuring emf because it

(1) is portable **(2)** draws no current at balance

(3) has low resistance **(4)** is cheaper

34. The magnetic force on a moving charge does no work because it is

(1) perpendicular to the velocity **(2)** parallel to the velocity

(3) zero always **(4)** constant

35. In a moving coil galvanometer, the magnetic field is made

(1) uniform **(2)** zero

(3) radial **(4)** axial

36. Which is diamagnetic?

(1) iron **(2)** aluminium

(3) nickel **(4)** bismuth

37. Lenz's law is a consequence of conservation of

(1) charge **(2)** energy

(3) momentum **(4)** mass

38. The rms value of an alternating current of peak value I_0 is

(1) $I_0/\sqrt{2}$ **(2)** $\sqrt{2}\, I_0$

(3) $I_0/2$ **(4)** I_0

39. The power factor of a purely inductive ac circuit is

(1) 1 **(2)** 0.5

(3) 0 **(4)** ∞

40. The speed of electromagnetic waves in vacuum is

(1) $\sqrt{(\mu_0\varepsilon_0)}$ **(2)** $\mu_0\varepsilon_0$

(3) $1/\mu_0\varepsilon_0$ **(4)** $1/\sqrt{(\mu_0\varepsilon_0)}$

41. An object is at 30 cm from a concave mirror of focal length 15 cm. The image is formed at

(1) 15 cm, erect **(2)** 30 cm, inverted

(3) 60 cm, inverted **(4)** infinity

42. Dispersion of white light by a prism occurs because the refractive index

(1) depends on wavelength **(2)** is constant

(3) depends on intensity **(4)** is less than 1

43. In photoelectric emission, the maximum kinetic energy of photoelectrons depends on

(1) intensity of light **(2)** area of the plate

(3) frequency of light **(4)** time of exposure

44. In the Bohr model, the radius of the nth orbit is proportional to

(1) n **(2)** 1/n

(3) $1/n^2$ **(4)** n^2

45. A Zener diode is mainly used as a

(1) rectifier **(2)** voltage regulator

(3) amplifier **(4)** oscillator

Chemistry

46. The molar mass of H_2SO_4 is

(1) 98 g mol^{-1} **(2)** 96 g mol^{-1}

(3) 49 g mol^{-1} **(4)** 100 g mol^{-1}

47. The empirical formula of glucose ($C_6H_{12}O_6$) is

(1) $C_6H_{12}O_6$ **(2)** CHO

(3) CH_2O **(4)** $C_2H_4O_2$

48. When 2 g of H_2 reacts with 32 g of O_2 to form water, the limiting reagent is

(1) O_2 **(2)** H_2

(3) both **(4)** neither

49. The orbital with n = 3, l = 1 is

(1) 3s **(2)** 3d

(3) 2p **(4)** 3p

50. According to the aufbau principle, 4s is filled before 3d because 4s has

(1) lower (n + l) value **(2)** higher energy

(3) more electrons **(4)** larger l

51. The element with Z = 17 belongs to group

(1) 7 **(2)** 1

(3) 17 **(4)** 15

52. The element with the most negative electron gain enthalpy is

(1) F **(2)** Cl

(3) Br **(4)** I

53. An ionic bond is most likely between

(1) C and H **(2)** N and O

(3) Cl and Cl **(4)** Na and Cl

54. The shape of NH_3 is

(1) trigonal pyramidal **(2)** trigonal planar

(3) tetrahedral **(4)** T-shaped

55. The hybridisation of carbon in ethyne is

(1) sp^3 **(2)** sp^2

(3) sp **(4)** dsp^2

56. The number of lone pairs on oxygen in H_2O is

(1) 1 **(2)** 2

(3) 3 **(4)** 0

57. At constant temperature, the pressure of a fixed mass of gas is inversely proportional to its volume. This is

(1) Charles' law **(2)** Avogadro's law

(3) Dalton's law **(4)** Boyle's law

58. Which is a state function?

(1) internal energy **(2)** work

(3) heat **(4)** path length

59. The enthalpy of neutralisation of a strong acid by a strong base is about

(1) $-13.7\ kJ\ mol^{-1}$ **(2)** $-5.7\ kJ\ mol^{-1}$

(3) $-57.1\ kJ\ mol^{-1}$ **(4)** $+57.1\ kJ\ mol^{-1}$

60. For $N_2 + 3H_2 \rightleftharpoons 2NH_3$, increasing the pressure

(1) shifts it backward **(2)** shifts it forward

(3) has no effect **(4)** stops the reaction

61. The conjugate base of HSO_4^- is

(1) H_2SO_4 **(2)** H_2O

(3) OH^- **(4)** SO_4^{2-}

62. For AgCl with solubility s mol L^{-1}, Ksp equals

(1) s^2 **(2)** s

(3) $4s^3$ **(4)** 2s

63. Oxidation is

(1) gain of electrons **(2)** loss of protons

(3) loss of electrons **(4)** gain of hydrogen

64. The oxidation number of S in H_2SO_4 is

(1) +4 **(2)** +6

(3) −2 **(4)** +2

65. Heavy water is used in nuclear reactors as a

(1) fuel **(2)** coolant only

(3) shield **(4)** moderator

66. Which alkali metal reacts most vigorously with water?

(1) Cs **(2)** Na

(3) Li **(4)** K

67. The formula of Plaster of Paris is

(1) $CaSO_4 \cdot 2H_2O$ **(2)** $CaSO_4$

(3) $CaSO_4 \cdot \frac{1}{2}H_2O$ **(4)** $CaCO_3$

68. BF_3 behaves as a

(1) Lewis base **(2)** Lewis acid

(3) Brønsted base **(4)** reducing agent

69. The allotrope of carbon that conducts electricity is

(1) diamond **(2)** charcoal (amorphous)

(3) fullerene C_{60} **(4)** graphite

70. The IUPAC name of CH_3CH_2CHO is

(1) propanal **(2)** propanone

(3) propan-1-ol **(4)** propanoic acid

71. The group showing a strong $-I$ effect is

(1) $-CH_3$ **(2)** $-C_2H_5$

(3) $-NO_2$ **(4)** $-OCH_3$ (+R dominant)

72. Propene reacts with HBr (no peroxide) to give mainly

(1) 1-bromopropane **(2)** 2-bromopropane

(3) 1,2-dibromopropane **(4)** propane

73. Benzene mainly undergoes

(1) nucleophilic addition **(2)** free-radical addition

(3) elimination **(4)** electrophilic substitution

74. The major greenhouse gas among these is

(1) CO_2 **(2)** N_2

(3) O_2 **(4)** Ar

75. The unit of molality is

(1) mol L^{-1} **(2)** g L^{-1}

(3) mol kg^{-1} **(4)** mol

76. By Raoult's law, the relative lowering of vapour pressure equals the

(1) molality **(2)** mole fraction of solute

(3) mole fraction of solvent **(4)** molarity

77. Osmotic pressure of a dilute solution is given by

(1) $\pi = RT/C$ **(2)** $\pi = C/RT$

(3) $\pi = nR/T$ **(4)** $\pi = CRT$

78. The standard hydrogen electrode has a potential of

(1) 0 V **(2)** 1 V

(3) -1 V **(4)** 0.76 V

79. One faraday is approximately

(1) 9650 C **(2)** 1 C

(3) 96500 C **(4)** 6.022×10^{23} C

80. For a zero-order reaction, the rate

(1) doubles when concentration doubles **(2)** is independent of concentration

(3) depends on square of concentration **(4)** is always zero

81. A catalyst increases the rate of a reaction by

(1) increasing ΔH **(2)** increasing temperature

(3) shifting equilibrium **(4)** lowering activation energy

82. Transition metals show variable oxidation states because

(1) (n−1)d and ns energies are close **(2)** they are metals

(3) they have high density **(4)** they form alloys

83. The most common oxidation state of lanthanoids is

(1) +2 **(2)** +4

(3) +3 **(4)** +1

84. The IUPAC name of $[Co(NH_3)_5Cl]Cl_2$ is

(1) chloridopentaamminecobalt(II) chloride **(2)** pentaamminechloridocobalt(III) chloride

(3) pentaamminecobalt(III) chloride **(4)** cobalt pentaammine dichloride

85. An SN2 reaction at a chiral carbon proceeds with

(1) racemisation **(2)** retention

(3) no change **(4)** inversion of configuration

86. Chloroform is stored in dark, filled bottles to prevent formation of

(1) phosgene **(2)** chlorine

(3) methane **(4)** carbon tetrachloride

87. Phenol is more acidic than ethanol because

(1) it has more H atoms **(2)** it is aromatic only

(3) phenoxide ion is resonance-stabilised **(4)** it has a higher molar mass

88. Tollens' reagent can distinguish

(1) alcohols from ethers **(2)** aldehydes from ketones

(3) alkenes from alkynes **(4)** acids from esters

89. Amines are basic because of

(1) the C–N bond **(2)** hydrogen bonding

(3) their smell **(4)** the lone pair on nitrogen

90. Which base is found in RNA but not in DNA?

(1) uracil **(2)** thymine

(3) adenine **(4)** guanine

Biology — Botany

91. Binomial nomenclature was introduced by

(1) Darwin **(2)** Linnaeus

(3) Whittaker **(4)** Mendel

92. The basic unit of classification is
(1) species **(2)** genus
(3) family **(4)** order

93. The five-kingdom classification was proposed by
(1) Linnaeus **(2)** Haeckel
(3) Whittaker **(4)** Copeland

94. The cell wall of fungi is made of
(1) cellulose **(2)** peptidoglycan
(3) pectin **(4)** chitin

95. Bacteria without a cell wall are
(1) cyanobacteria **(2)** mycoplasma
(3) actinomycetes **(4)** spirochaetes

96. Viroids contain only
(1) free RNA without a protein coat **(2)** DNA with a protein coat
(3) protein **(4)** RNA with a protein coat

97. A lichen is an association between
(1) bacteria and fungi **(2)** algae and bacteria
(3) algae and fungi **(4)** fungi and roots

98. Agar is commercially obtained from
(1) Chlorella **(2)** Spirogyra
(3) Laminaria **(4)** Gelidium

99. The amphibians of the plant kingdom are
(1) pteridophytes **(2)** bryophytes
(3) gymnosperms **(4)** algae

100. In pteridophytes, the main plant body is the
(1) sporophyte **(2)** gametophyte
(3) protonema **(4)** prothallus

101. Gymnosperms are characterised by
(1) enclosed seeds **(2)** absence of seeds
(3) naked seeds **(4)** double fertilisation

102. Double fertilisation is characteristic of
(1) bryophytes **(2)** pteridophytes
(3) gymnosperms **(4)** angiosperms

103. Whorled phyllotaxy is seen in
(1) china rose **(2)** Alstonia
(3) mustard **(4)** sunflower

104. The primary (tap) root develops from the
(1) radicle **(2)** plumule
(3) hypocotyl **(4)** cotyledon

105. Epipetalous stamens are found in
(1) mustard **(2)** pea
(3) brinjal **(4)** lily

106. Potato, tomato and belladonna belong to the family

(1) Fabaceae **(2)** Liliaceae
(3) Brassicaceae **(4)** Solanaceae

107. In a monocot stem, vascular bundles are

(1) arranged in a ring and open **(2)** scattered and closed
(3) radial **(4)** absent

108. Secondary growth is generally absent in monocots due to absence of

(1) cambium **(2)** xylem
(3) phloem **(4)** pith

109. Annual rings are formed due to activity of

(1) cork cambium **(2)** apical meristem
(3) vascular cambium **(4)** intercalary meristem

110. The cell theory was proposed by

(1) Watson and Crick **(2)** Robert Hooke
(3) Singer and Nicolson **(4)** Schleiden and Schwann

111. Ribosomes of prokaryotes are of the type

(1) 80S **(2)** 70S
(3) 60S **(4)** 40S

112. The fluid mosaic model of the plasma membrane was proposed by

(1) Singer and Nicolson **(2)** Schleiden and Schwann
(3) Watson and Crick **(4)** Robertson

113. The 'powerhouse of the cell' is the

(1) nucleus **(2)** Golgi body
(3) mitochondrion **(4)** ribosome

114. The light reactions of photosynthesis occur in the

(1) stroma **(2)** cytoplasm
(3) outer membrane **(4)** thylakoids (grana)

115. The longest phase of the cell cycle is

(1) M phase **(2)** interphase
(3) cytokinesis **(4)** anaphase

116. Crossing over occurs during

(1) pachytene **(2)** leptotene
(3) diakinesis **(4)** zygotene

117. Meiosis results in

(1) doubling of chromosomes **(2)** no change in chromosome number
(3) halving of chromosome number **(4)** polyploidy

118. The water potential of pure water at atmospheric pressure is

(1) positive **(2)** negative
(3) infinite **(4)** zero

119. Stomatal opening is associated with

(1) loss of K^+ from guard cells **(2)** influx of K^+ into guard cells
(3) increase in CO_2 **(4)** abscisic acid

120. Biological nitrogen fixation is catalysed by

(1) nitrogenase **(2)** nitrate reductase
(3) RuBisCO **(4)** hydrogenase

121. The primary CO_2 acceptor in C_3 plants is

(1) PEP **(2)** OAA
(3) RuBP **(4)** PGA

122. Which is a C_4 plant?

(1) wheat **(2)** rice
(3) potato **(4)** maize

123. Photolysis of water occurs in association with

(1) PS I **(2)** PS II
(3) stroma **(4)** mitochondria

124. The end product of glycolysis is

(1) pyruvate **(2)** acetyl CoA
(3) glucose **(4)** citrate

125. The Krebs cycle takes place in the

(1) cytoplasm **(2)** inner membrane only
(3) mitochondrial matrix **(4)** chloroplast

126. The respiratory quotient of fats is about

(1) 1 **(2)** 4
(3) zero **(4)** 0.7

127. The hormone that promotes fruit ripening is

(1) auxin **(2)** ethylene
(3) cytokinin **(4)** gibberellin

128. Bolting in rosette plants is caused by

(1) gibberellins **(2)** auxins
(3) ABA **(4)** ethylene

129. The pigment that perceives photoperiod is

(1) chlorophyll **(2)** carotene
(3) phytochrome **(4)** xanthophyll

130. The endosperm of angiosperms is generally

(1) haploid **(2)** diploid
(3) tetraploid **(4)** triploid

131. Formation of seeds without fertilisation is called

(1) parthenocarpy **(2)** apomixis
(3) polyembryony **(4)** vegetative propagation

132. The phenotypic ratio of a Mendelian monohybrid F_2 is

(1) 3 : 1 **(2)** 1 : 2 : 1
(3) 9 : 3 : 3 : 1 **(4)** 1 : 1

133. In snapdragon flower colour, the F_2 phenotypic ratio is 1 : 2 : 1 because of

(1) dominance **(2)** epistasis
(3) incomplete dominance **(4)** linkage

134. The inducer of the lac operon is

(1) glucose **(2)** galactose
(3) tryptophan **(4)** lactose (allolactose)

135. The ecological pyramid that is always upright is the pyramid of

(1) numbers **(2)** energy
(3) biomass **(4)** standing crop

Biology — Zoology

136. Radial symmetry is found in

(1) Cnidaria **(2)** Annelida
(3) Arthropoda **(4)** Chordata

137. The notochord is a characteristic of

(1) annelids **(2)** molluscs
(3) chordates **(4)** arthropods

138. True metameric segmentation is first seen in

(1) Platyhelminthes **(2)** Nematoda
(3) Mollusca **(4)** Annelida

139. A water vascular system is found in

(1) Porifera **(2)** Echinodermata
(3) Mollusca **(4)** Cnidaria

140. The unique feature of mammals is

(1) mammary glands **(2)** four-chambered heart
(3) internal fertilisation **(4)** lungs

141. The blood of cockroach is colourless because it lacks

(1) plasma **(2)** WBCs
(3) haemoglobin **(4)** water

142. The heart of a frog is

(1) two-chambered **(2)** four-chambered
(3) one-chambered **(4)** three-chambered

143. The inner lining of the trachea has

(1) squamous epithelium **(2)** ciliated epithelium
(3) cuboidal epithelium **(4)** keratinised epithelium

144. A ligament connects

(1) bone to bone **(2)** muscle to bone
(3) muscle to muscle **(4)** nerve to muscle

145. The simplest amino acid is

(1) alanine **(2)** serine
(3) glycine **(4)** valine

146. Most enzymes are

(1) lipids **(2)** carbohydrates
(3) nucleic acids **(4)** proteins

147. Salivary amylase acts on

(1) proteins **(2)** starch
(3) fats **(4)** nucleic acids

148. Bile helps in

(1) emulsification of fats **(2)** digestion of proteins
(3) absorption of water **(4)** digestion of starch

149. An increase in CO_2 shifts the oxygen–haemoglobin dissociation curve to the right. This is the

(1) Haldane effect **(2)** Hamburger shift
(3) Bohr effect **(4)** chloride shift

150. The tidal volume of a healthy adult is about

(1) 1200 mL **(2)** 2500 mL
(3) 3000 mL **(4)** 500 mL

151. The universal donor blood group is

(1) AB^+ **(2)** O^-
(3) A^+ **(4)** B^-

152. The pacemaker of the human heart is the

(1) SA node **(2)** AV node
(3) bundle of His **(4)** Purkinje fibres

153. Normal human blood pressure is about

(1) 140/100 mm Hg **(2)** 90/60 mm Hg
(3) 120/80 mm Hg **(4)** 160/90 mm Hg

154. The structural and functional unit of the kidney is the

(1) neuron **(2)** ureter
(3) loop of Henle **(4)** nephron

155. ADH (vasopressin) promotes

(1) glucose reabsorption **(2)** water reabsorption
(3) sodium excretion **(4)** urine formation increase

156. Muscle contraction is explained by sliding of

(1) actin and myosin **(2)** keratin and collagen
(3) actin and tubulin **(4)** myosin and elastin

157. Gout is caused by accumulation of

(1) urea **(2)** calcium
(3) uric acid crystals **(4)** cholesterol

158. Saltatory conduction occurs in

(1) unmyelinated fibres **(2)** cardiac muscle
(3) smooth muscle **(4)** myelinated nerve fibres

159. The resting membrane potential of a neuron is about

(1) +70 mV **(2)** −70 mV
(3) 0 mV **(4)** −7 mV

160. Insulin is secreted by

(1) β-cells of islets of Langerhans **(2)** α-cells
(3) δ-cells **(4)** acinar cells

161. Synthesis of thyroxine requires

(1) iron **(2)** calcium
(3) iodine **(4)** zinc

162. The 'fight or flight' hormone is

(1) insulin **(2)** thyroxine
(3) oxytocin **(4)** adrenaline

163. Spermatogenesis takes place in the

(1) epididymis **(2)** seminiferous tubules
(3) vas deferens **(4)** prostate

164. Ovulation is triggered by a surge of

(1) LH **(2)** FSH
(3) progesterone **(4)** prolactin

165. In humans, fertilisation usually occurs in the

(1) uterus **(2)** cervix
(3) ampullary region of the fallopian tube **(4)** ovary

166. The blastocyst implants in the

(1) myometrium **(2)** perimetrium
(3) cervix **(4)** endometrium

167. Copper-T is an example of

(1) a barrier method **(2)** an intra-uterine device
(3) an oral pill **(4)** a surgical method

168. Down syndrome is caused by

(1) trisomy 21 **(2)** monosomy X
(3) trisomy 18 **(4)** XXY

169. Haemophilia is

(1) autosomal dominant **(2)** autosomal recessive
(3) X-linked recessive **(4)** Y-linked

170. The double helix model of DNA was proposed by

(1) Meselson and Stahl **(2)** Hershey and Chase
(3) Griffith **(4)** Watson and Crick

171. Semi-conservative DNA replication was proved by

(1) Watson and Crick **(2)** Meselson and Stahl
(3) Avery **(4)** Nirenberg

172. The start codon AUG codes for

(1) methionine **(2)** valine
(3) tryptophan **(4)** leucine

173. The human haploid genome has about

(1) 3×10^6 bp **(2)** 3×10^{12} bp
(3) 3×10^9 bp **(4)** 3×10^3 bp

174. The theory of natural selection was proposed by

(1) Lamarck **(2)** de Vries
(3) Mendel **(4)** Darwin

175. Forelimbs of whale, bat and human are examples of

(1) analogous organs **(2)** homologous organs

(3) vestigial organs **(4)** convergent evolution

176. Malaria is transmitted by the female

(1) Anopheles **(2)** Culex

(3) Aedes **(4)** housefly

177. Vaccination provides

(1) passive immunity **(2)** innate immunity

(3) active immunity **(4)** no immunity

178. HIV mainly infects

(1) B-lymphocytes **(2)** RBCs

(3) platelets **(4)** helper T-lymphocytes

179. Restriction enzymes recognise

(1) random sequences **(2)** palindromic sequences

(3) only RNA **(4)** only proteins

180. The insecticidal protein of Bt cotton comes from

(1) Bacillus thuringiensis **(2)** Agrobacterium tumefaciens

(3) E. coli **(4)** Rhizobium

Answer Key

Physics

1. (2)	**2.** (1)	**3.** (3)	**4.** (4)	**5.** (2)	**6.** (1)
7. (3)	**8.** (4)	**9.** (2)	**10.** (1)	**11.** (3)	**12.** (4)
13. (2)	**14.** (1)	**15.** (3)	**16.** (4)	**17.** (2)	**18.** (1)
19. (3)	**20.** (4)	**21.** (2)	**22.** (1)	**23.** (3)	**24.** (4)
25. (2)	**26.** (1)	**27.** (3)	**28.** (4)	**29.** (2)	**30.** (1)
31. (3)	**32.** (4)	**33.** (2)	**34.** (1)	**35.** (3)	**36.** (4)
37. (2)	**38.** (1)	**39.** (3)	**40.** (4)	**41.** (2)	**42.** (1)
43. (3)	**44.** (4)	**45.** (2)			

Chemistry

46. (1)	**47.** (3)	**48.** (2)	**49.** (4)	**50.** (1)	**51.** (3)
52. (2)	**53.** (4)	**54.** (1)	**55.** (3)	**56.** (2)	**57.** (4)
58. (1)	**59.** (3)	**60.** (2)	**61.** (4)	**62.** (1)	**63.** (3)
64. (2)	**65.** (4)	**66.** (1)	**67.** (3)	**68.** (2)	**69.** (4)
70. (1)	**71.** (3)	**72.** (2)	**73.** (4)	**74.** (1)	**75.** (3)
76. (2)	**77.** (4)	**78.** (1)	**79.** (3)	**80.** (2)	**81.** (4)
82. (1)	**83.** (3)	**84.** (2)	**85.** (4)	**86.** (1)	**87.** (3)
88. (2)	**89.** (4)	**90.** (1)			

Biology — Botany

91. (2)	**92.** (1)	**93.** (3)	**94.** (4)	**95.** (2)	**96.** (1)
97. (3)	**98.** (4)	**99.** (2)	**100.** (1)	**101.** (3)	**102.** (4)
103. (2)	**104.** (1)	**105.** (3)	**106.** (4)	**107.** (2)	**108.** (1)
109. (3)	**110.** (4)	**111.** (2)	**112.** (1)	**113.** (3)	**114.** (4)
115. (2)	**116.** (1)	**117.** (3)	**118.** (4)	**119.** (2)	**120.** (1)
121. (3)	**122.** (4)	**123.** (2)	**124.** (1)	**125.** (3)	**126.** (4)
127. (2)	**128.** (1)	**129.** (3)	**130.** (4)	**131.** (2)	**132.** (1)
133. (3)	**134.** (4)	**135.** (2)			

Biology — Zoology

136. (1)	**137.** (3)	**138.** (4)	**139.** (2)	**140.** (1)	**141.** (3)
142. (4)	**143.** (2)	**144.** (1)	**145.** (3)	**146.** (4)	**147.** (2)
148. (1)	**149.** (3)	**150.** (4)	**151.** (2)	**152.** (1)	**153.** (3)
154. (4)	**155.** (2)	**156.** (1)	**157.** (3)	**158.** (4)	**159.** (2)
160. (1)	**161.** (3)	**162.** (4)	**163.** (2)	**164.** (1)	**165.** (3)
166. (4)	**167.** (2)	**168.** (1)	**169.** (3)	**170.** (4)	**171.** (2)
172. (1)	**173.** (3)	**174.** (4)	**175.** (2)	**176.** (1)	**177.** (3)
178. (4)	**179.** (2)	**180.** (1)			

Original model paper prepared for practice, following the latest announced exam pattern. Not an official NTA paper.